Customer Churn and Retention Analysis for an E-commerce Platform

- A comprehensive SQL case study

Vishal Agrawal

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1.0 Business Problem

- Customer churn is a major concern for e-commerce and subscription-based businesses, as acquiring new customers is often more expensive than retaining existing ones.
- This project aims to analyze historical transaction data to identify behavioral patterns that indicate when customers are likely to stop purchasing or disengage from the platform.
- Understanding these patterns can help businesses implement targeted strategies to improve customer retention.

2.0 Project Objective

The objective of this project is to analyze customer transaction and activity data to better understand customer retention patterns and identify factors associated with churn.

Key business goal:

Provide data-driven insights and recommendations that help improve customer retention and reduce customer churn.

3.0 Dataset Description

The dataset used in this project is the **Online Retail II Dataset**.

Original file:

Online_retail_II.xlsx

Dataset characteristics:

- ~525,000 transactions
- 8 columns representing transactional sales data
- Time period covering approximately one year of online retail activity

The dataset contains transaction-level data including order details, product information, purchase quantity, pricing, customer identifiers, and geographic location.

4.0 Data Preparation for MySQL

Several preprocessing steps were performed before importing the dataset into **MySQL**.

1. The dataset contained missing customer IDs for many rows, which likely represent guest purchases. These rows were removed using **pandas** since churn analysis requires identifiable customers.

2. Column names were standardized by converting them to lowercase and replacing spaces with underscores to ensure SQL compatibility.

3. The customer_id column was converted from the default float data type to an integer.

4. Because the original dataset contained over 500K rows, which could slow down query performance during development, a random sample of **100,000 transactions** was created.

The cleaned and sampled dataset was then imported into MySQL for further analysis and SQL-based data cleaning.

5.0 Data Import and Configuration in MySQL

The SQL queries for this project were written and executed using the DBeaver SQL editor.

A database connection was established with the MySQL server, and a new project database named **ecommerce_churn_analysis** was created to store the project tables.

The processed **Online Retail dataset** containing approximately **100,000 transaction records** was then imported into MySQL as the **primary raw data table (sql_project)**.

This table served as the **source dataset for all subsequent data cleaning, transformation, and schema design steps** performed during the project.

6.0 Data Cleaning in MySQL

Real-world datasets often contain inconsistencies and invalid records. Therefore, several data validation and cleaning steps were performed using SQL queries.

• The dataset was first checked for rows with **missing customer_id values**. Since these rows were already removed during the preprocessing stage in Python, the query returned **zero missing values**.

• A small number of records were found where the **same customer_id was associated with different countries**. These inconsistent records were identified and removed to maintain data integrity.

• The dataset contained **381 duplicate transaction records** where the combination of **invoice number, product ID (stock_code), quantity, and unit price** was repeated. These duplicates were detected using aggregation queries and removed to avoid double counting in analysis.

• Approximately **2,250 rows contained cancelled orders**, identified by invoice numbers beginning with **"C"**. These rows also contained **negative quantity values**, representing returned items. Since the focus of the project is customer purchasing behavior, these records were removed.

• The dataset also contained **11 records with a unit price of 0**, which indicates invalid or incomplete transactions. These rows were removed from the dataset.

After these steps, the dataset contained **clean and reliable transaction records suitable for further analysis**.

7.0 Database Schema Design

To organize the transactional data into a structured relational format, the cleaned dataset was transformed into a **normalized database schema**.

Four tables were created from the main transaction dataset:

- **Customers** – Stores unique customer information such as **customer_id** and **country.**
- **Products** – Contains product-level information including **stock_code** and **product description.**
- **Orders** – Stores order-level details such as **invoice_no**, **customer_id**, and **order date**.
- **Order Items** – Represents the transactional relationship between **orders** and **products**, including **quantity purchased** and **unit price.**

Each table was created with an appropriate **primary key**, and relationships between tables were maintained using **foreign keys** to ensure referential integrity.

This schema design improves **query efficiency, data organization, and scalability for analytical queries** performed later in the project.

8.0 Populating Data into Schema Tables

Data from the original transaction table was normalized and inserted into the relational schema tables (customers, products, orders, and order_items). During this process, several data consistency issues such as duplicate keys and inconsistent values were addressed.

Customers Table

- The **customers** table was populated using the **customer_id** and **country** fields from the source table.
- Since some **customer_id** values were associated with multiple countries in the raw dataset, a single country value was selected for each customer to maintain the uniqueness of the primary key and prevent duplicate records.

Products Table

- The **products** table was created using the **stock_code** and **description** columns from the source data.
- These fields were renamed **product_id** and **product_name** to follow standard naming conventions.
- In cases where multiple descriptions existed for the same **stock_code**, only one description was retained to ensure a single record per product.

Orders Table

- The **orders** table was populated using the **invoice, customer_id, and invoice_date** fields from the source dataset.
- The columns were renamed to **order_id** and **order_date** to align with conventional order management terminology.
- Since multiple **invoice_date** values were occasionally recorded for the same invoice, the earliest recorded date was selected for each order.

Order Items Table

- The **order_items** table was populated using transaction-level data including **invoice, stock_code, quantity, and price,** along with an auto-incremented **order_item_id** as the primary key.
- Column names were standardized to match the schema structure.
- During the insertion process, it was observed that some product prices contained very small values (e.g., 0.001). The initial schema defined the **price** column as DECIMAL (10,2),

which caused these values to be truncated to 0. To preserve pricing accuracy, the column definition was modified to DECIMAL (10,3).

9.0 SQL Based Exploratory Data Analysis

9.1 Overview of Dataset

To get an idea of the scale of the dataset queries were executed on the schema tables and following information was obtained

Metric	Value
Number Of Customers	4147
Number Of Orders	16413
Number Of Products	3648
Number Of Order Items	97252
Total Revenue Generated	2,138,331

9.2 Monthly Orders and Monthly Revenue Trend

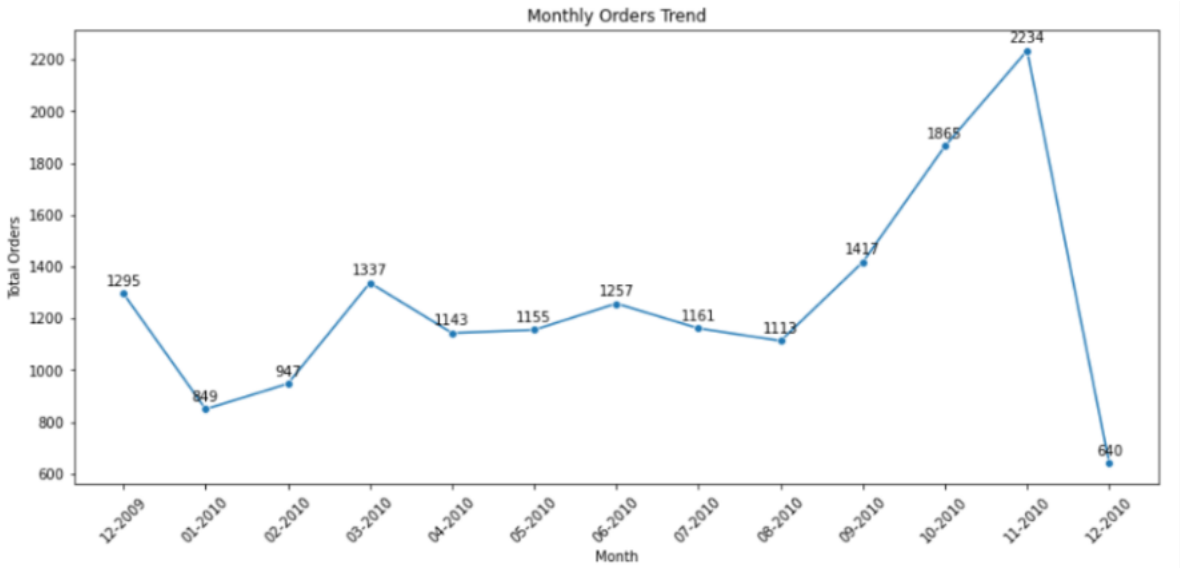
Business Questions:

1. Are there seasonal patterns in customer order behavior across the year?
2. How does total monthly revenue fluctuate over time, and are there identifiable peak and low-performing periods?

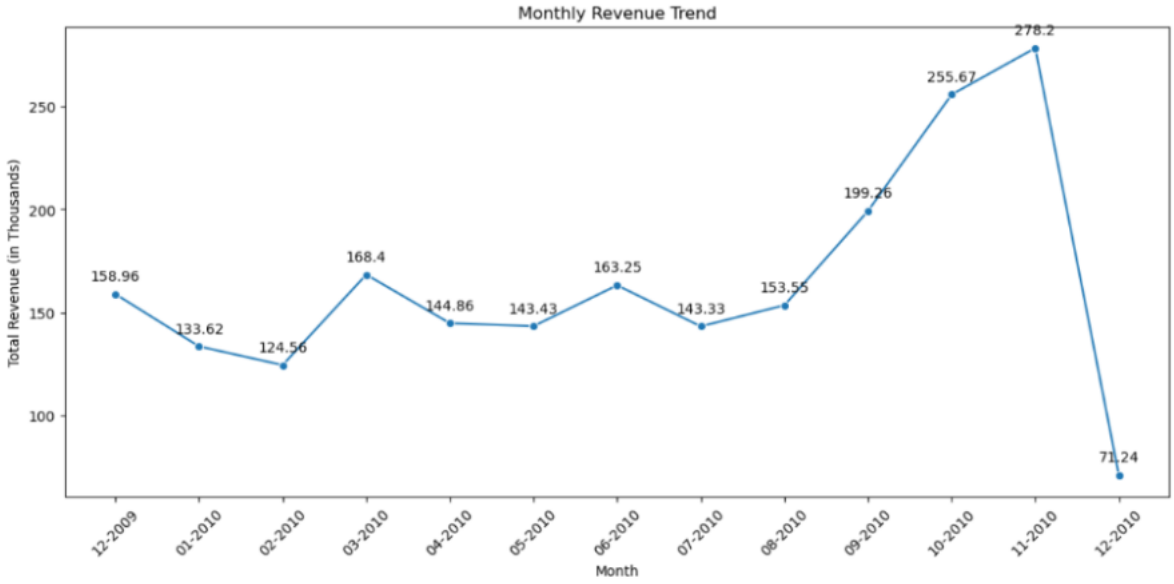
Analysis: The number of orders placed, and the total revenue generated per month were calculated using SQL queries, and the results were analyzed for spotting trends and seasonality by plotting line plots in Python.

Supporting Visualizations:

Monthly Orders Trend



Monthly Revenue Trend



Key Insights

From the above visualizations, it can be observed that both **order volume and total revenue increase significantly during the months of September, October, and November**. This pattern suggests the presence of **seasonal demand, likely driven by holiday or year-end shopping periods**.

In contrast, the remaining months show relatively **stable ordering behavior with moderate fluctuations in both orders and revenue**. The sharp decline observed in December 2010 is likely due to **incomplete data for that month in the dataset**.

These trends highlight the importance of **seasonality in customer purchasing behavior**, which will be further explored in later sections while analyzing **customer retention and churn patterns**.

9.3 Customer Purchase Frequency Analysis

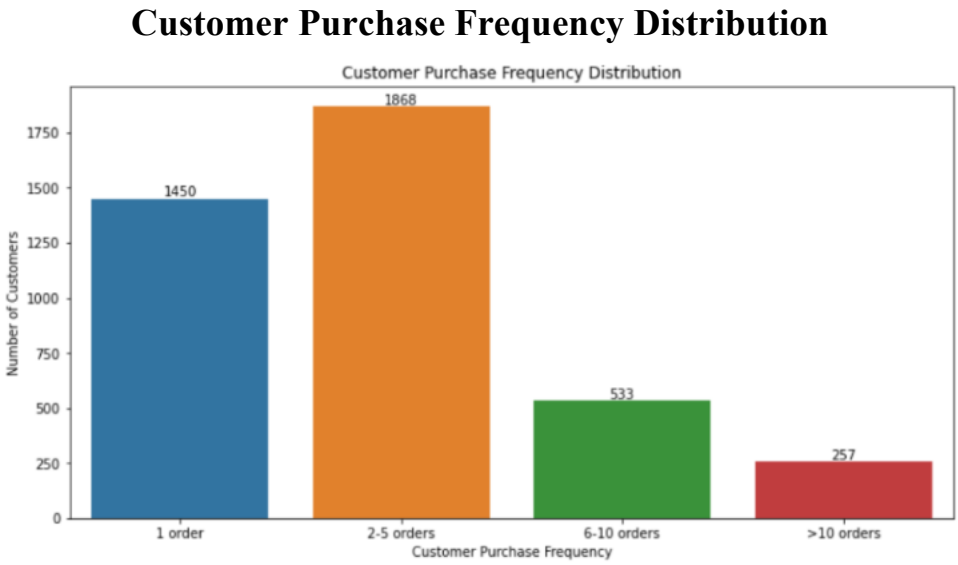
Business Question: How frequently do customers place orders, and what proportion of customers are repeat buyers?

Analysis: To better understand customer purchasing behavior, customers were segmented based on the total number of orders they placed during the observed period. Segmenting customers by purchase frequency helps identify engagement patterns and provides an initial view of potential customer retention and churn behavior, which will be explored in later sections of the analysis.

Customers were grouped into the following categories based on their order count:

- 1 order
- 2–5 orders
- 6–10 orders
- More than 10 orders

Supporting Visualization:



Key Insights

The **2–5 orders segment contains the highest number of customers**, indicating that a large portion of customers place multiple orders but remain within a relatively low purchase frequency range. In contrast, the number of customers placing **more than 6 orders declines significantly**, suggesting that fewer customers transition into higher-frequency purchasing behavior.

This pattern may indicate a **drop-off in repeat purchasing after the initial few orders**, which could potentially signal early-stage customer churn. Further retention and cohort analysis will be conducted in subsequent sections to investigate this behavior in more detail.

9.4 Customer Lifetime Value Analysis

Business Question: What is the distribution of Customer Lifetime Value (CLV), and to what extent do high-value customers drive overall revenue?

Analysis: To identify the most valuable customers, a customer lifetime value (CLV) analysis was conducted using transaction data. Since the dataset represents a limited time period rather than the full customer lifecycle, total revenue generated by each customer was used as a proxy for customer lifetime value.

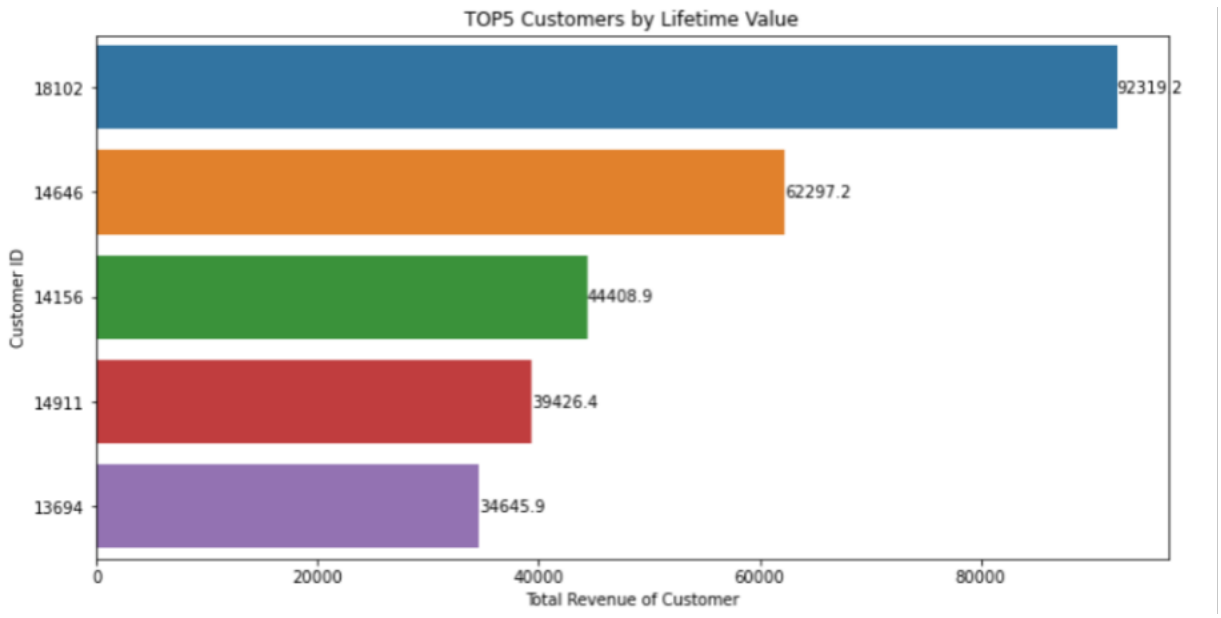
The analysis calculated the following metrics for each customer:

- **Total Orders** – total number of orders placed by the customer
- **Total Customer Revenue** – total revenue generated by the customer
- **Average Order Value** – average revenue per order for each customer

The query aggregated revenue at the order level and then summarized it at the customer level to ensure accurate calculation of order values

Supporting Visualization:

TOP 5 customers by Lifetime Value (Total Revenue Generated)



Key Insights

The analysis reveals that a **small group of customers contributes to a disproportionately large share of total revenue**. These high-value customers typically exhibit:

- Higher purchase frequency
- Larger average order values
- Consistent repeat purchases

This pattern highlights the importance of **retaining high-value customers**, as losing even a small number of them could significantly impact overall revenue.

Identifying such customers will help the platform implement targeted **loyalty programs, personalized promotions, and retention strategies**.

9.5 Product Sales Distribution Analysis

Business Question: Do a small number of products contribute disproportionately to overall revenue?

Analysis: To understand which products drive customer purchases, product-level sales analysis was conducted using transaction data. Two key metrics were analyzed:

- **Total quantity sold per product**
- **Total revenue generated per product**

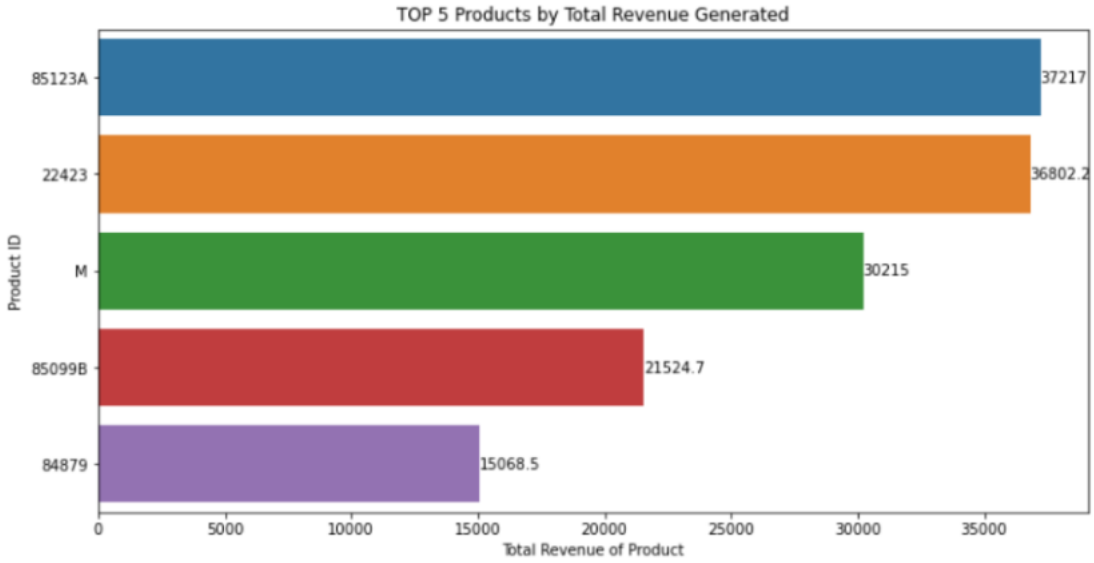
Two queries were executed to identify:

- **Top-selling products by quantity**
- **Top revenue-generating products**,

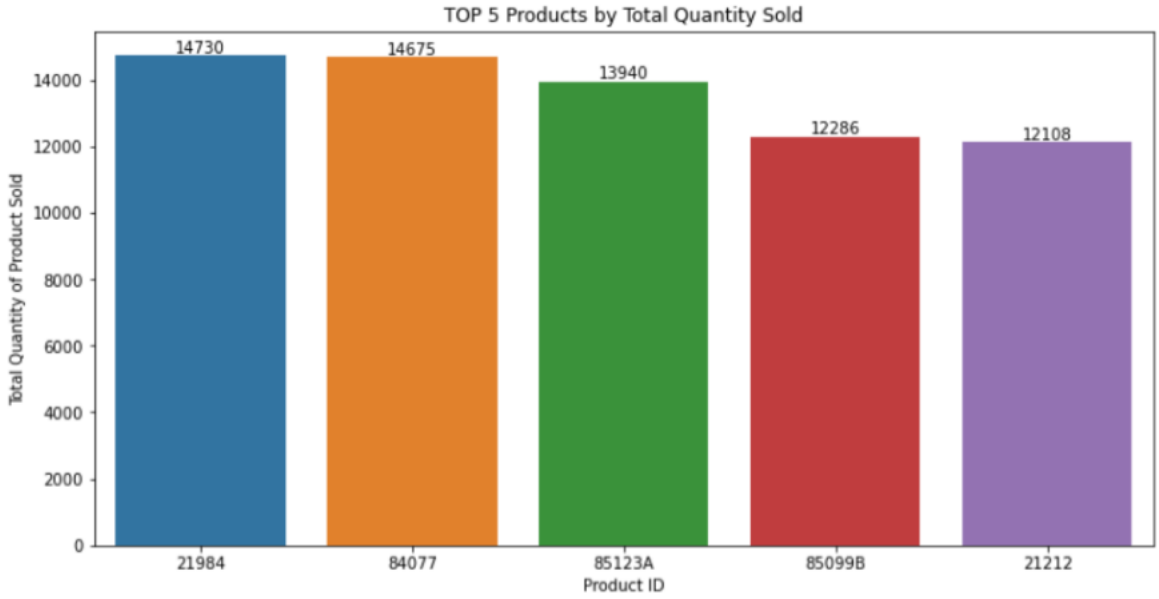
The results reveal the products that play the most significant role in driving customer purchases and overall revenue.

Supporting Visualizations:

TOP 5 Products by Total Revenue Generated



TOP 5 Products by Quantity Sold



Key Insights

The analysis shows that a **small subset of products contributes disproportionately to total sales volume and revenue**. Some products achieve high sales volumes due to frequent purchases, while others generate high revenue despite lower quantities due to higher prices.

This indicates that product demand is **not evenly distributed**, with certain products acting as key drivers of customer transactions.

Understanding these product-level patterns is useful for:

- Inventory planning
- Product promotion strategies
- Identifying products that encourage repeat purchases

9.6 Average Order Value (AOV) Analysis

Business Question: What is the average order value (AOV), and what does it reveal about customer spending patterns on the platform?

Analysis: To evaluate customer purchasing behavior, the Average Order Value (AOV) across all orders was calculated. The distribution of order values was also analyzed to understand the variation in customer spending and identify how different order sizes contribute to overall revenue.

Summary Statistics:

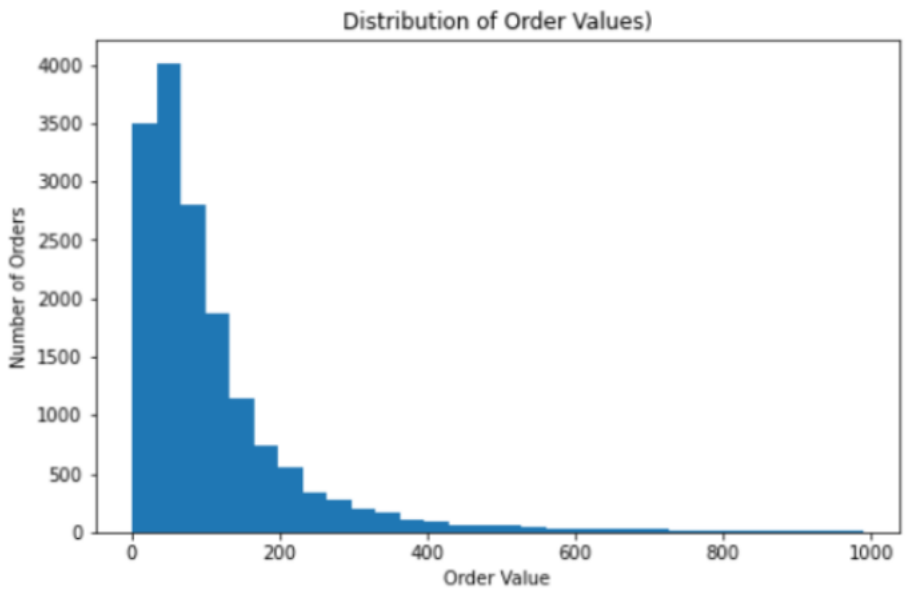
Metric	Value
Minimum Order Value	0.42
Average Order Value	130.28
Maximum Order Value	13032.26

Order Value Distribution:

Order Category	Number of Orders
High Value Orders	3565
Medium Value Orders	7259
Low Value Orders	5589

Supporting Visualization:

Histogram of Order Values



Key Insights

- The **Average Order Value (AOV) is 130.28**, indicating a moderate typical spend per transaction, with most customer purchases falling within lower-value ranges.

- The **histogram of order values shows that the majority of transactions fall within the 0–500 range**, suggesting that most purchases are relatively small, while the number of orders declines sharply as order value increases.

- The distribution is **right-skewed**, with a small number of very high-value orders (up to 13,032.26) contributing disproportionately to overall revenue.

Business Recommendations

Implement strategies such as **product bundling, upselling, and targeted promotions** to increase Average Order Value, while identifying and retaining high-spending customers through personalized marketing and loyalty initiatives.

9.7 Customer Geographic Distribution Analysis

Business Question: How are customers distributed across countries, and which regions contribute the most to overall revenue?

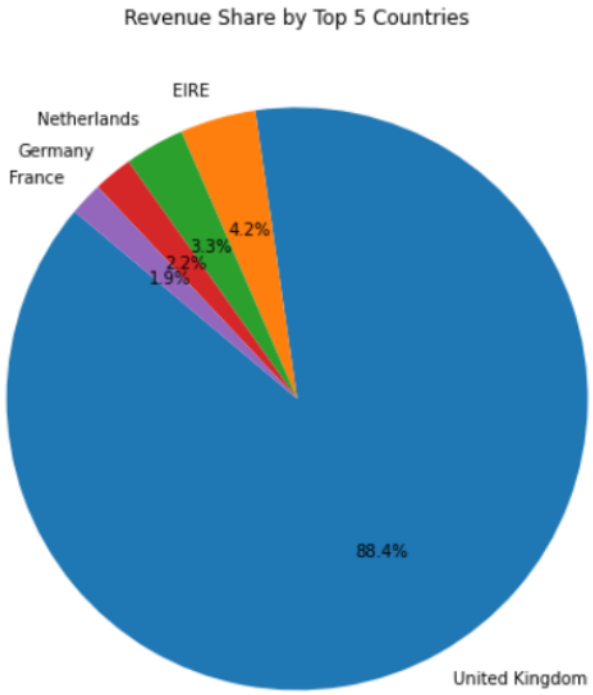
Analysis: To understand the geographic distribution of customers and identify key markets, the number of customers, total revenue generated, and total orders were analyzed across different countries

Results Summary:

Country	Number Of Customers	Total Revenue	Total Orders
United Kingdom	3779	1792494.965	15077
EIRE	3	85490.2	263
Netherlands	21	67530.61	104
Germany	65	44189.751	291
France	46	37567.93	209

Supporting Visualization:

Pie Chart of TOP 5 Countries by Revenue share percentage



Key Insights

- The **United Kingdom dominates the customer base and revenue**, accounting for the vast majority of customers, orders, and sales, indicating that the business is heavily concentrated in the UK market.

- **EIRE generates exceptionally high revenue from only a few customers**, suggesting the presence of highly valuable customers with large order sizes or frequent purchases.

- **European countries such as the Netherlands, Germany, and France represent important secondary markets**, contributing consistent orders and revenue beyond the UK.

- While customers exist across multiple global regions, **non-European countries contribute relatively small revenue and order volumes**, indicating limited market penetration outside core European markets.

Business Recommendations

Strengthen customer acquisition and marketing efforts in **high-potential international markets (particularly Europe)** while exploring targeted expansion strategies in underpenetrated regions to diversify revenue beyond the UK market.

9.8 Basket Size Analysis

Business Question: What is the distribution of basket sizes across orders, and how many items do customers typically purchase per transaction?

Analysis: Basket size represents the total number of items purchased in each order and provides insight into customer purchasing behavior. It was calculated by summing the quantity of items within each order. Additionally, the average basket size and the distribution of basket sizes were analyzed by categorizing orders into three basket size groups.

Summary Statistics:

Metric	Value
Minimum Basket Size	1
Average Basket Size	80.5
Maximum Basket Size	25200

Results Summary:

Basket Category	Number of Orders
Small Baskets (1-5 items)	1161
Medium Baskets (6-15 items)	2705
Large Baskets (>15 items)	12547

Extremely Large Orders:

Order Id	Items in Order
495194	25200
502269	20000
518505	15783
497946	15216
524174	11664

Key Insights

- The **average basket size is 80.5 items**, indicating that customers often purchase multiple items per transaction, suggesting a strong tendency toward bulk or multi-item purchases.

- **Large baskets (>15 items) dominate order activity**, accounting for the majority of transactions (12,547 orders), while small baskets (1–5 items) represent only a small share of total orders.

- A small number of **extremely large orders (over 10,000 items)** exist in the dataset, indicating potential wholesale purchases, bulk buyers, or business customers contributing significantly to item volume.

Business Recommendations

Identify and segment **high-volume customers responsible for large basket purchases**, and develop targeted strategies such as bulk discounts, loyalty programs, or dedicated B2B offerings to strengthen relationships and maximize revenue from these high-value buyers.

10.0 Customer Churn and Retention Analysis

Definition of Churn

Since the dataset covers **only about 1 year**, churn must be **defined by inactivity.**

For the Analysis below, A customer is considered churned if they have not placed an order in the last **90 days** of the dataset

Project KPIs Summary:

Metric	Value
Average Order Value	130.28
Average Purchase Interval (Days)	74.78
Average Basket Size (items per order)	80.5

10.1 Customer Churn Rate Analysis

Business Question: What proportion of customers stop purchasing over time, and how significant is customer churn within the platform?

Analysis: Based on the churn definition described above, customers were categorized into churned and retained groups. The churn rate was then calculated as the percentage of customers who churned relative to the total customer base.

Results Summary

Metric	Value
Total Customers	4147
Churned Customers	1368
Churn Rate	32.99
Retention Rate	67.01

Key Insights

- Approximately **32.99% of customers are classified as churned** based on the 90-day inactivity threshold, indicating that nearly one-third of the customer base has stopped purchasing.
- In absolute terms, **1,368 customers have churned**, representing a substantial potential loss of future revenue if these customers are not re-engaged.
- While **67.01% of customers remain active**, the relatively high churn proportion suggests that customer retention is a critical area for improving long-term revenue stability.

Business Recommendations

Implement **targeted retention strategies for customers approaching the inactivity threshold**, such as personalized promotions, re-engagement campaigns, or loyalty incentives, to reduce churn and preserve long-term customer value

10.2 Customer Retention Rate Analysis

Business Question: What percentage of customers become repeat buyers after their first purchase, and how strong is customer retention on the platform?

Analysis: To assess customer retention behavior, customers were segmented based on purchase frequency into two groups: single-purchase customers and repeat customers. The retention rate was then calculated as the proportion of repeat customers relative to the total customer base

Results Summary

Metric	Value
Customers with 1 order	1450
Customers with multiple orders	2658
Retention Percentage	64.70

Key Insights

- **64.70% of customers made repeat purchases**, indicating that a majority of customers return to the platform after their initial transaction, reflecting a moderate level of customer retention.
- However, **35.30% of customers placed only a single order**, highlighting a sizable group of one-time buyers who do not progress into repeat purchasing behavior.

Business Recommendations

Implement targeted **post-purchase engagement strategies**—such as follow-up promotions, personalized product recommendations, or loyalty incentives—to encourage first-time buyers to make additional purchases and improve overall customer retention.

10.3 Customer Purchase Interval Analysis

Business Question: What are the typical time intervals between customer purchases, and how frequently do customers return to place new orders?

Analysis: To understand how frequently customers return to make purchases, the number of days between consecutive orders for each customer was calculated using SQL window functions. These purchase intervals were then categorized into three groups to analyze customer buying frequency:

Frequent Buyers: less than 30 days between purchases
Moderate Buyers: 30–90 days between purchases
Rare Buyers: more than 90 days between purchases

Customer Purchase Frequency Distribution

Category	Number of Customers
Rare Buyers (> 90 days between purchases)	**733**
Moderate Buyers (30–90 days)	**1343**
Frequent Buyers (< 30 days)	**582**

Statistics Summary

Metric	Value (Days)
Highest days between two purchases	364
Lowest days between two purchases	0
Highest average days between purchases (per customer)	362
Lowest average days between purchases	0
Average purchase interval across customers	74.78

Key Insights

- **Moderate buyers (30–90 days between purchases) represent the largest segment with 1,343 customers**, indicating that most customers return within a few months after their previous purchase.
- **Frequent buyers (<30 days) account for 582 customers**, representing a highly engaged group that purchases regularly and likely contributes significantly to repeat revenue.
- **A sizable segment of customers (733) are rare buyers**, taking more than 90 days between purchases, which may indicate lower engagement and a higher likelihood of churn.
- Purchase behavior varies widely across customers, with **some placing multiple orders on the same day**, while others return only after long gaps (up to **364 days**), highlighting diverse purchasing patterns across the customer base.

Business Recommendations

- Develop **targeted retention and engagement strategies for rare buyers**, such as personalized promotions, reminder campaigns, or loyalty incentives to encourage more frequent purchases.
- Strengthen relationships with **frequent buyers through loyalty programs, exclusive offers, or early product access** to retain these high-value customers and maximize repeat revenue.

Additional Observation

- Out of **4,147 total customers**, **1,489 customers were excluded from the purchase frequency analysis** because they had zero or only one recorded order, making it impossible to calculate a purchase interval. These customers may represent **new customers, one-time buyers, or churned users**, and should be further examined in churn and retention analyses

10.4 Relationship Between Purchase Frequency and Churn

Business Question: Are customers who purchase more frequently less likely to churn compared to those with fewer orders?

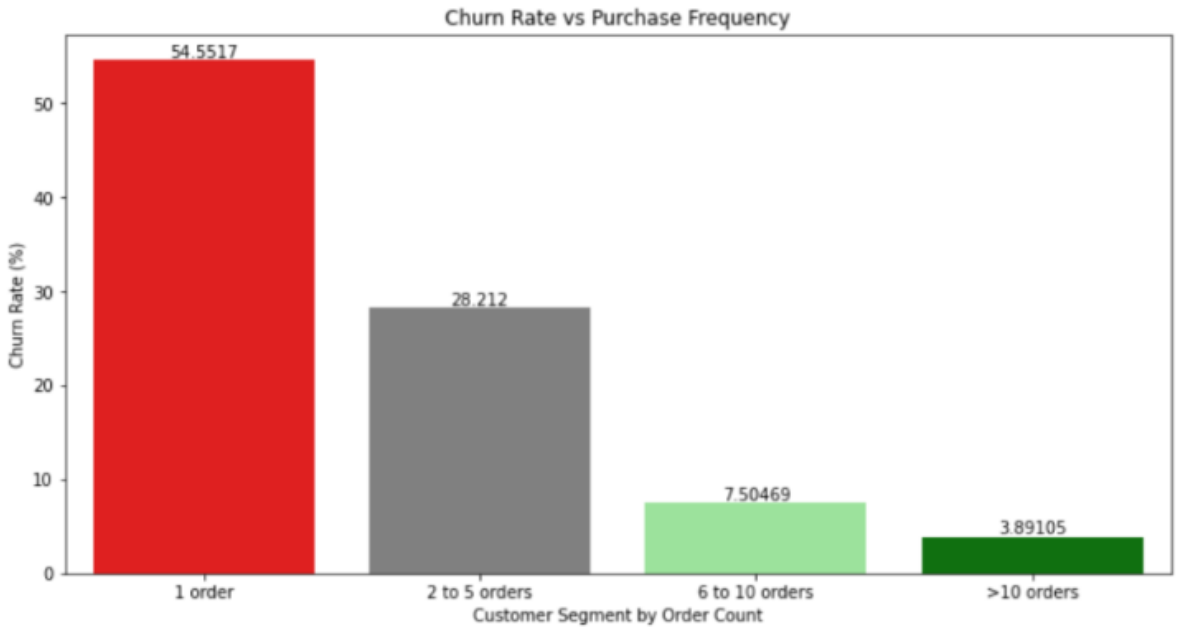
Analysis: To analyze the relationship between purchase frequency and customer churn, customers were segmented into four groups based on the total number of orders placed (1 order, 2–5 orders, 6–10 orders, and more than 10 orders). The churn rate was then calculated for each segment to assess how purchasing activity impacts customer retention.

Results Summary

Customer Segment	Total Customers	Churn Customers	Churn Rate	Retention Rate
1 order	1450	791	54.55%	45.45%
2–5 orders	1868	527	28.21%	71.79%
6–10 orders	533	40	7.50%	92.50%
>10 orders	257	10	3.89%	96.11%

Supporting Visualization:

Churn Rate Vs Purchase Frequency



Key Insights

- **Customers with only one order exhibit the highest churn rate (54.55%)**, indicating that more than half of first-time buyers do not return for a second purchase, making this the most vulnerable stage of the customer lifecycle.

- **Churn drops significantly once customers begin making repeat purchases**, with customers placing 2–5 orders showing a much lower churn rate of **28.21%**, suggesting that early repeat purchases greatly improve retention.

- **Highly engaged customers demonstrate very strong loyalty**, with churn rates falling to **7.50% for customers with 6–10 orders** and only **3.89% for customers with more than 10 orders**, indicating that frequent buyers are highly stable and valuable.

- The **bar chart clearly illustrates a strong inverse relationship between purchase frequency and churn**, showing that churn probability decreases consistently as the number of customer orders increases.

Business Recommendations

- Prioritize **early-stage retention strategies** aimed at converting first-time buyers into repeat customers, such as post-purchase follow-ups, targeted promotions, or personalized recommendations.
- Strengthen engagement with **high-frequency customers through loyalty programs, exclusive offers, and personalized incentives** to maintain long-term relationships with these highly valuable customers.

10.5 Customer Churn Analysis by Customer Value Segment

Business Question: Are high-value customers less likely to churn compared to low-value customers?

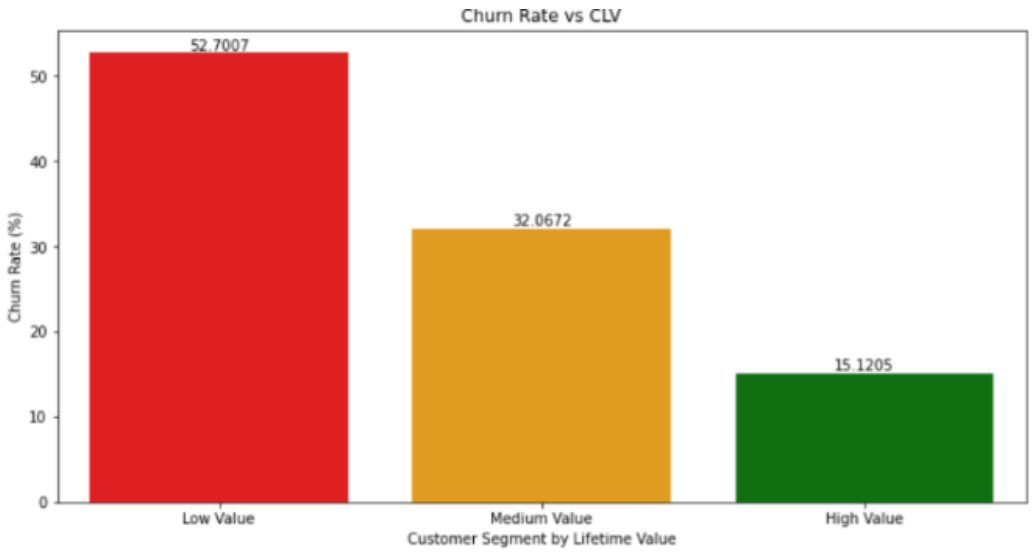
Analysis: To analyze how customer value influences churn behavior, customers were segmented into **Low, Medium, and High Customer Lifetime Value (CLV) groups** based on total revenue generated per customer. Revenue-based tertiles were used to create balanced segments. Churn rates were then calculated for each segment using a **90-day inactivity threshold**.

Results Summary

Customer Value Segment	Total Customers	Churn Customers	Churn Rate	Retention Rate
Low Value	1370	722	52.70%	47.30%
Medium Value	1369	439	32.07%	67.93%
High Value	1369	207	15.12%	84.88%

Supporting Visualization:

Churn Rate Vs Customer Lifetime Value (Bar Chart)



Key Insights

• Customer Lifetime Value is strongly linked to retention

There is a clear inverse relationship between customer value and churn. The churn rate declines sharply from **52.7% for low-value customers to 15.1% for high-value customers**, indicating that higher-value customers tend to maintain longer relationships with the business.

• Low-value customers represent the largest churn risk segment

More than half of customers in the low-value segment become inactive, suggesting weaker engagement and lower long-term commitment. This segment contributes disproportionately to the overall churn.

• High-value customers demonstrate strong loyalty and stability

With a **retention rate of nearly 85%**, high-value customers show consistent purchasing behavior and long-term engagement, making them the most reliable revenue contributors.

• Bar chart confirms a clear retention gradient across value segments

The bar chart visually highlights a steady decline in churn rates from low to high value segments, reinforcing the strong inverse relationship between **customer lifetime value and churn probability**.

Business Recommendations

• Protect high-value customers through targeted retention strategies

Implement loyalty programs, VIP benefits, and personalized engagement initiatives to maintain strong relationships with high-value customers whose retention has the greatest impact on revenue stability.

• Improve engagement among low and medium-value customers

Introduce targeted promotions, personalized marketing, and repeat-purchase incentives to increase purchase frequency and gradually move customers into higher lifetime value segments.

Overall, the findings indicate that **customer lifetime value is a key predictor of churn behavior**, suggesting that value-based segmentation can be an effective framework for designing targeted retention and revenue growth strategies.

10.6 Customer Lifetime Value Before Churn

Business Question: How much revenue is lost due to customer churn, and which customer value segments drive the largest financial impact?

Analysis: To assess the financial impact of customer churn, the **average revenue generated by customers prior to churn** was calculated. Churned customers were then segmented into four revenue-based groups (Low, Medium, High, and Very High) to evaluate how revenue loss is distributed across customer value tiers and to identify which segments contribute most to lost revenue.

Summary Statistics

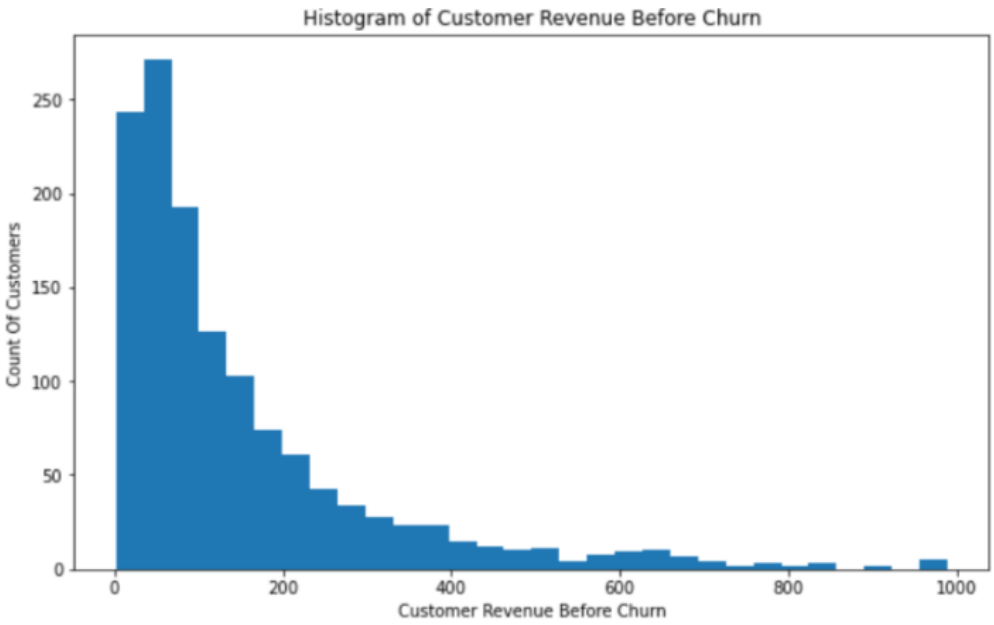
Metric	Value
Average Revenue Before Churn	212.1
Maximum Revenue Before Churn	8702
Minimum Revenue Before Churn	1.65

Distribution of Churned Customers by Revenue Segment

Revenue Segment	Number of Churned Customers
Low Revenue (<100)	707
Medium Revenue (100–500)	554
High Revenue (500–1000)	68
Very High Revenue (>1000)	39

Supporting Visualization:

Histogram of Customer Revenue Before Churn



Key Insights

- **Churn is concentrated among low-spending customers.**
  The majority of churned customers belong to the **Low (707) and Medium (554) revenue segments**, indicating that lower-value customers are significantly more likely to disengage from the platform.

- **Average revenue before churning is relatively modest compared to extreme values.**
  The **average revenue before churning is 212.1**, while the **maximum reaches 8702**, showing that a small subset of customers contributes disproportionately higher revenue before leaving.

- **Revenue distribution among churned customers is highly skewed.**
  The **histogram shows a strong right-skewed distribution**, confirming that most churned customers generate relatively low revenue, while only a small number of customers account for very high spending before churning.

- **High-value churn is rare but financially impactful.**
  Only **68 High-Revenue and 39 Very-High-Revenue customers** churned, but losing even a small number of these customers can lead to **significant revenue loss due to their large individual contribution**.

Business Recommendations

- **Increase engagement among low-value customers to grow lifetime value.**
  Since most churn occurs among low- and medium-revenue customers, targeted promotions, onboarding improvements, and repeat-purchase incentives can help convert these customers into more loyal and higher-value segments.

- **Prioritize proactive retention for high-value customers.**
  Implement **VIP programs, personalized offers, and early churn detection strategies** for high-spending customers to protect the small but highly valuable segment that contributes disproportionately to total revenue.

10.7 RFM Customer Segmentation Analysis

Business Question: Which customer segments contribute the most value to the business and which segments show early signs of disengagement requiring targeted retention strategies?

Analysis: To better understand customer behavior and engagement levels, customers were segmented using **RFM Analysis**, which evaluates customers based on three dimensions: **Recency (how recently a customer made a purchase), Frequency (how often they purchase), and Monetary value (how much they spend)**.

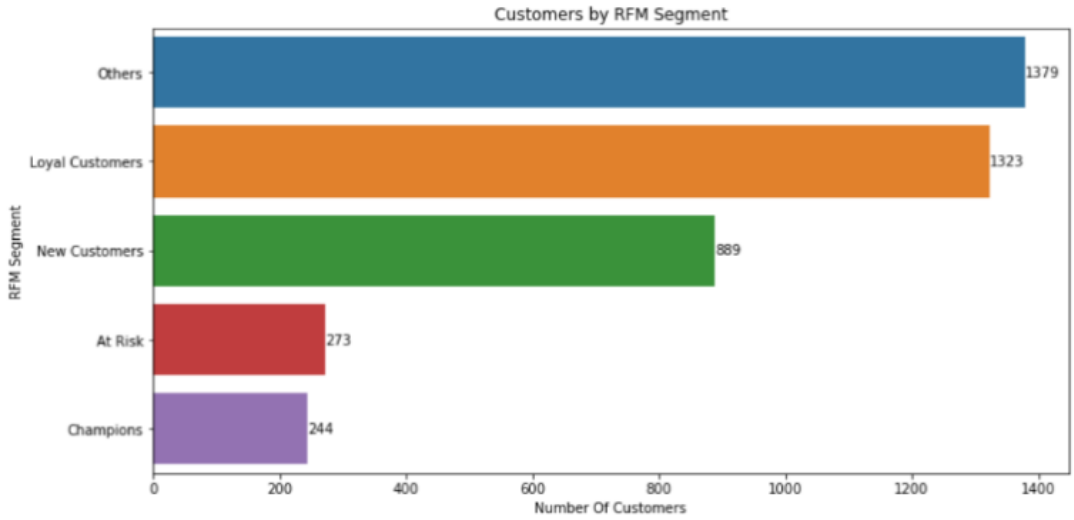
Customers were assigned RFM scores using SQL window functions and grouped into behavioral segments such as **Champions, Loyal Customers, New Customers, At Risk, and Others**

Results Summary

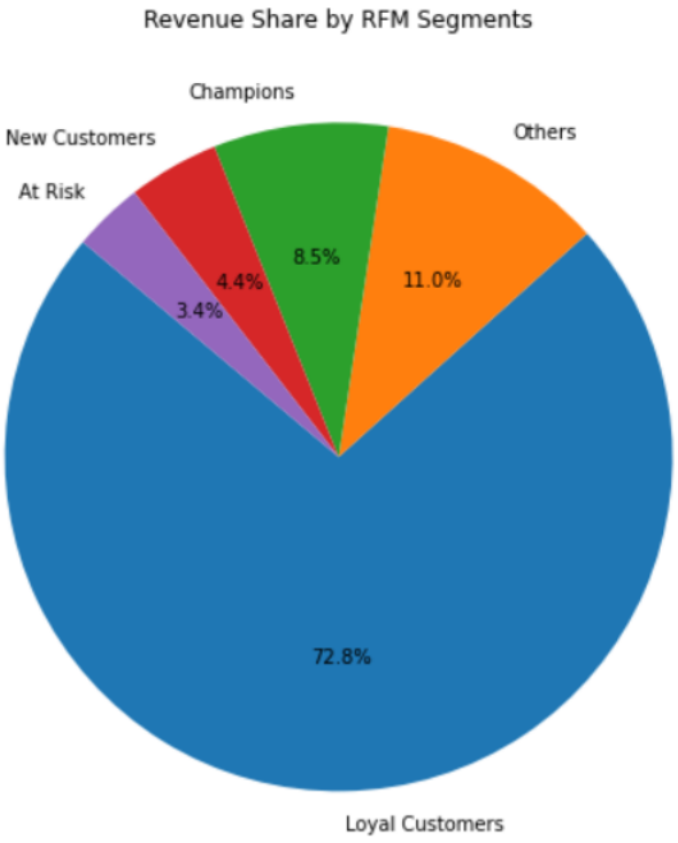
Segment	Customers
Champions	244
Loyal Customers	1323
At Risk	273
New Customers	889
Others	1379

Supporting Visualizations:

Number Of Customers by RFM segment



RFM Segments by Revenue Share



Key Insights

- **Loyal Customers represent the core revenue drivers**, accounting for **72.76% of total revenue despite representing only 1323 customers**, highlighting their critical importance to overall business performance.
- **Champions (244 customers) are the most valuable segment**, demonstrating high recency, frequency, and spending behavior. These customers show the strongest engagement and contribute disproportionately to revenue.
- **New Customers (889 customers) represent strong future growth potential**, as their recent purchase activity suggests successful acquisition, but their long-term value depends on converting them into repeat buyers.
- **At Risk Customers (273 customers) show declining engagement**, indicating that previously active customers may be drifting toward churn without targeted intervention.
- **A large portion of customers fall into the "Others" segment (1379 customers)**, reflecting mixed or inconsistent purchasing behavior and suggesting opportunities for further segmentation and engagement strategies

Business Recommendations

- **Prioritize retention of high-value segments (Champions and Loyal Customers)** through loyalty programs, personalized offers, and exclusive benefits to protect the primary revenue base.

- **Implement targeted lifecycle marketing strategies**, including onboarding campaigns for New Customers and win-back promotions for At Risk Customers, to increase repeat purchases and reduce potential churn.

The RFM segmentation reveals that a relatively small portion of highly engaged customers (Champions and Loyal Customers) contributes a disproportionate share of revenue, highlighting that **customer retention and loyalty-focused strategies are significantly more impactful for revenue growth than acquisition alone.**

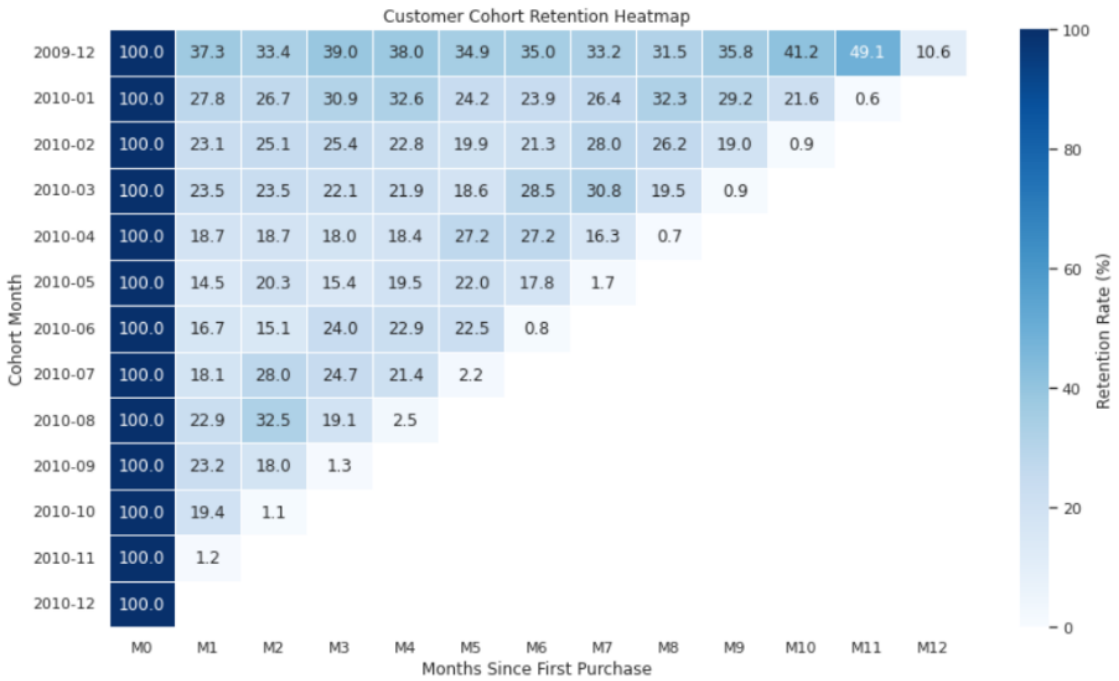
10.8 Cohort Retention Analysis

Business Question: How does customer retention change over time for customers acquired in different months?

Analysis: Cohort analysis was performed to understand how customer retention evolves over time by **grouping customers based on the month of their first purchase**. Each cohort represents customers acquired in the same month, and their purchasing activity was **tracked over subsequent months to evaluate retention behavior.**

Supporting Visualization:

Customer Cohort Retention Heatmap



Key Insights

- **A significant drop in retention after the first month:** Across nearly all cohorts, retention declines sharply from 100% in Month 0 to roughly **18–37% by Month 1**, indicating that a large proportion of customers do not return shortly after their first purchase.
- **Retention stabilizes after the early lifecycle stage:** Following the initial drop, most cohorts stabilize between **20–35% retention from months 3–10**, suggesting the presence of a consistent base of repeat customers who continue purchasing over time.
- **Heatmap reveals a classic ecommerce retention pattern:** The cohort retention heatmap clearly shows **high initial churn followed by a stable long-term customer base**, indicating that long-term revenue is driven primarily by a smaller group of loyal repeat buyers.
- **Evidence of delayed repeat purchases:** Some cohorts show **retention increases in later months** (e.g., the 2009-12 cohort rising again around months 9–11), suggesting that customers may return after longer purchase intervals rather than buying consistently every month.
- **Newer cohorts show lower early retention but limited observation window:** Recent cohorts appear to have weaker retention in early months, though this is likely due to **shorter observation periods and incomplete lifecycle data**, meaning their long-term retention patterns may still evolve.

Business Recommendations

- **Focus retention strategies on the first 30–60 days:** Since the majority of churn occurs shortly after the first purchase, targeted onboarding strategies such as follow-up emails, second-purchase discounts, and personalized recommendations could significantly improve early retention.
- **Implement structured re-engagement campaigns:** Because some customers return after longer gaps, reminder emails, personalized promotions, and product recommendations can encourage customers to return sooner and increase purchase frequency.
- **Strengthen loyalty and repeat-purchase incentives:** Developing loyalty programs or rewards for repeat purchases can help retain the stable core group of customers and increase overall customer lifetime value.

The cohort analysis reveals that **customer churn occurs predominantly within the first month after acquisition**, while a smaller core group of repeat customers drives long-term engagement. This highlights that **improving early customer experience and post-purchase engagement could significantly increase overall retention and lifetime value.**

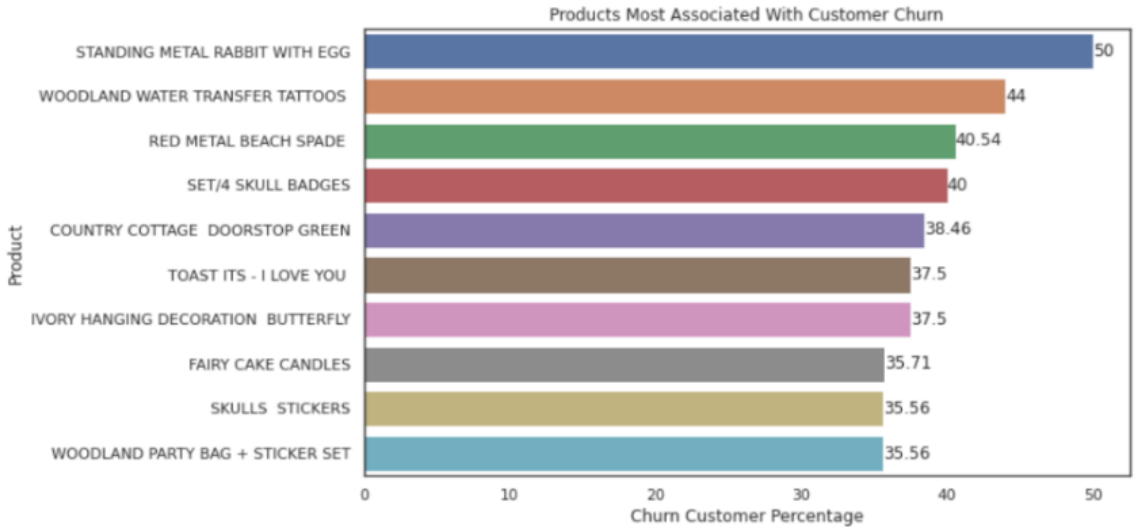
10.9 Product Behavior of Churned Customers

Business Question: Are certain products associated with higher customer churn rates, indicating potential product-level drivers of customer disengagement

Analysis: To examine whether customer churn is associated with specific products, customers were first classified as **active or churned** using a 90-day inactivity threshold. For each product, the number of purchases made by active and churned customers was calculated, and the **churn rate per product** was computed as the proportion of churned customers relative to the total customers who purchased that product. To ensure statistical reliability and avoid noise from low-volume items, the analysis was restricted to products with **at least 20 unique purchasing customers**.

Supporting Visualization:

TOP 10 Products associated with Churn



Key Insights

- **Certain products exhibit noticeably higher churn rates:** The bar chart shows churn percentages ranging from **35% to 50%**, with products such as *Standing Metal Rabbit with Egg* and *Woodland Water Transfer Tattoos* having the highest churn associations.
- **Many high-churn products are novelty or low-commitment items:** Products like stickers, badges, candles, and decorative accessories tend to be **impulse or one-time purchases**, which naturally results in lower repeat purchasing behavior.
- **These products may attract casual or first-time buyers:** Items such as novelty decorations and small gift products are often purchased for **specific occasions or events**, suggesting that customers who buy them may not have strong long-term engagement with the platform.
- **Moderate churn levels suggest correlation rather than causation:** While churn rates are relatively elevated for these products, the **35–50% range indicates they are likely early indicators of lower engagement customers rather than direct causes of churn.**

Business Recommendations

- **Implement targeted post-purchase retention campaigns:** Customers purchasing these products should receive **follow-up marketing such as personalized recommendations, reminder emails, or second-purchase discounts** to encourage repeat transactions.
- **Use cross-selling and bundling strategies:** Pair low-retention products with **higher-engagement or complementary items** to increase basket value and improve the likelihood that customers explore additional product categories.

The analysis suggests that several novelty and low-commitment products tend to attract **one-time or low-engagement buyers**, making them useful **early indicators of potential churn risk**. Targeted post-purchase engagement strategies can help convert these customers into repeat buyers and improve long-term retention.

11.0 SQL Techniques and Tools Used in Project

Key SQL Techniques Implemented

Data Aggregation

- GROUP BY
- Conditional Aggregation (COUNT)

Window Functions

- SUM (for calculating revenue share percentages of countries)
- NTILE (for segmenting customers based on revenue)
- ROW_NUMBER (for removing duplicate rows during data cleaning)
- LAG (for detecting interval between consecutive orders per customer)
- FIRST_VALUE (for computing retention rates for cohort analysis)

Data Filtering

- WHERE
- HAVING

JOINS

- Inner Join
- Left Join
- Cross Join

Data Transformation

- Common Table Expressions (CTEs)
- Subqueries
- Derived Tables
- CASE Statements

Date & Time Functions

- DATE_FORMAT
- DATEDIFF
- TIMESTAMPDIFF

Additional Functions

- COALESCE
- NULLIF
- ROUND
- DISTINCT

Tools Used in Project

- MySQL Server
- DBeaver
- Python (Pandas, Matplotlib, Seaborn)

12.0 Key SQL Queries

The following queries represent key analytical techniques used at various stages of the project

Data Cleaning

Removing duplicate rows using Window Functions

```sql
-- Add a column row_id for detecting duplicates
ALTER TABLE sql_project
ADD COLUMN row_id INT AUTO_INCREMENT PRIMARY KEY;

-- Create a Table with row_ids having duplicate data
CREATE TABLE duplicates AS
SELECT *
FROM (
    SELECT *,
            ROW_NUMBER() OVER (
                PARTITION BY invoice, stock_code, quantity, price
                ORDER BY row_id
            ) AS rn
    FROM sql_project
) t
WHERE rn > 1;

-- Verify the number of duplicates
SELECT COUNT(*) FROM duplicates;

-- Delete the duplicate rows from main table
DELETE FROM sql_project
WHERE row_id IN (
    SELECT row_id FROM duplicates
);
```

Exploratory Data Analysis

Order Value Distribution by High Value, Low Value, Medium Value Orders

```sql
WITH order_revenue AS
(
SELECT
    order_id,
    SUM(price * quantity) AS order_value
FROM ecommerce_churn_analysis.order_items
GROUP BY order_id
)

SELECT
    CASE
        WHEN order_value < 50 THEN 'Low Value'
        WHEN order_value BETWEEN 50 AND 150 THEN 'Medium Value'
        ELSE 'High Value'
    END AS order_value_segment,
    COUNT(*) AS number_of_orders
FROM order_revenue
GROUP BY order_value_segment;
```

Customer Purchase Frequency Analysis

```
WITH Customers_with_number_of_orders AS
(
SELECT c.customer_id, COUNT(o.order_id) AS Number_Of_Orders
FROM ecommerce_churn_analysis.customers c JOIN ecommerce_churn_analysis.orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id)

SELECT
COUNT(CASE WHEN Number_Of_Orders = 1 THEN 1 END)  AS Customers_with_1_order,
COUNT(CASE WHEN Number_Of_Orders BETWEEN 2 AND 5 THEN 1 END)  AS Customers_with_2_to_5_orders,
COUNT(CASE WHEN Number_Of_Orders BETWEEN 6 AND 10 THEN 1 END)  AS Customers_with_6_to_10_orders,
COUNT(CASE WHEN Number_Of_Orders > 10 THEN 1 END) AS Customers_with_more_than_10_orders
FROM Customers_with_number_of_orders;
```

Churn Analysis

Relationship Between customer revenue and churn behavior

```
WITH customer_lifetime_value AS (

SELECT o.customer_id, SUM(oi.price*oi.quantity) AS customer_revenue, MAX(o.order_date) AS last_order_date
FROM ecommerce_churn_analysis.order_items oi JOIN ecommerce_churn_analysis.orders o
ON oi.order_id = o.order_id
GROUP BY o.customer_id),
-- CTE for segmenting customers into revenue tertiles
customers_revenue_tertiles AS (

SELECT customer_id, customer_revenue,last_order_date,
      NTILE(3) OVER (ORDER BY customer_revenue) AS revenue_tertile
FROM customer_lifetime_value),

-- Assigning names to segments based on revenue
customer_value_segments AS (

SELECT customer_id, customer_revenue,last_order_date,

CASE
    WHEN revenue_tertile = 1                    THEN    'Low Value'
    WHEN revenue_tertile = 2                    THEN    'Medium Value'
    WHEN revenue_tertile = 3                    THEN    'High Value'
END AS customer_value_segment
FROM customers_revenue_tertiles),
```

```sql
-- Calculating the dataset end date to check for churn
dataset_end AS (
    SELECT MAX(order_date) AS dataset_end_date
    FROM ecommerce_churn_analysis.orders
)

-- Calculting total churn customers in each segment and the churn rate
SELECT
    customer_value_segment,
    COUNT(*) AS total_customers,
    COUNT(CASE WHEN DATEDIFF(d.dataset_end_date, c.last_order_date) > 90 THEN 1 END) AS churn_customers,
    COUNT(CASE WHEN DATEDIFF(d.dataset_end_date, c.last_order_date) > 90 THEN 1 END)*100.0/COUNT(*) AS churn_rate,
    100.0 - (COUNT(CASE WHEN DATEDIFF(d.dataset_end_date, c.last_order_date) > 90 THEN 1 END)*100.0/COUNT(*)) AS retention_rate
FROM customer_value_segments c
CROSS JOIN dataset_end d
GROUP BY customer_value_segment;
```

13.0 Summary of Key Insights and Business Recommendations

Summary of Key Insights

1. **The first purchase is the highest-risk point in the customer lifecycle.**
   More than half of customers who placed only one order churned (54.55%), compared to just 28.21% for customers with 2–5 orders. This 26-percentage-point drop between the first and second order represents the single largest retention opportunity in the entire dataset. The cohort heatmap reinforces this — across nearly all acquisition months, 63–82% of customers were lost within the first month after their initial purchase.

2. **A small group of customers drives the overwhelming majority of revenue.**
   RFM segmentation reveals that Loyal Customers alone account for 72.76% of total revenue despite representing only 1,323 of 4,147 customers. The top 5 customers by lifetime value generated between £34,000 and £92,000 each. This extreme revenue concentration means losing even a handful of high-value customers carries disproportionate financial consequences.

3. **Customer churn is substantial but unevenly distributed across value segments.**
   The overall churn rate stands at 32.99%, meaning nearly one in three customers has become inactive. However, churn is far from uniform — low-value customers churn at 52.70% while high-value customers churn at only 15.12%. This gradient confirms that value and loyalty are deeply linked, and that the majority of churn risk is concentrated in the lower-spending customer base.
4. **Purchasing behavior is strongly seasonal, with a concentrated revenue peak.**

Orders and revenue rise sharply from September through November, with November recording the highest order volume at 2,234 orders and £278,200 in revenue — nearly double the average of quieter months. Outside this window, monthly performance remains relatively flat, indicating that the business is heavily dependent on a short seasonal peak to drive annual results.

5. **RFM segmentation identifies 273 at-risk customers as the most recoverable churn opportunity.** These customers previously demonstrated high engagement but show recent declining activity. Unlike fully churned customers, at-risk customers still have an active relationship with the platform, making them significantly more responsive to re-engagement than cold win-back campaigns targeting long-inactive buyers.

6. **Certain product categories reliably signal early churn risk.** Products associated with the highest churn rates (35–50%) are predominantly novelty, low-commitment items — stickers, decorative accessories, and seasonal gifts. These products attract occasion-driven buyers who are unlikely to return without targeted intervention, making first-purchase product type a useful early predictor of churn probability.

Summary of Business Recommendations

1. **Launch a second-purchase conversion campaign as the highest-priority retention action.** Given that crossing from one order to two reduces churn by over 26 percentage points, a targeted post-purchase sequence — triggered within 7 days of a first order and offering a personalized discount or product recommendation — represents the highest-leverage intervention available. Even a modest improvement in first-to-second order conversion would materially reduce overall churn.

2. **Implement a pre-churn early warning system using the 74-day average purchase interval.** Rather than reacting after customers have already churned, flag customers who have been inactive for 60–75 days — before they cross the 90-day churn threshold — and trigger automated re-engagement campaigns. Acting within the window of likely return is significantly more effective than attempting to win back customers who have already gone cold.

3. **Protect high-value customers with a dedicated VIP retention program.** Champions and Loyal Customers collectively drive over 80% of the total revenue. This segment warrants dedicated retention investment including priority customer service, exclusive early access to new products, volume-based loyalty rewards, and personalized account management. The cost of retaining one high-value customer far outweighs the cost of acquiring a replacement.

4. **Target the 273 at-risk RFM customers with personalized win-back offers before they fully churn.** These customers represent the most cost-effective recovery opportunity — they have established purchase history, demonstrated prior loyalty, and have not yet fully disengaged. Campaigns referencing their specific purchase history and offering time-limited incentives are likely to outperform generic re-engagement efforts significantly.

5. **Integrate product-level churn signals into the post-purchase marketing workflow.** When a customer's first purchase falls into a high-churn product category (novelty items, seasonal gifts, low-commitment accessories), automatically enroll them in an extended onboarding sequence with cross-sell recommendations toward higher-engagement product categories. This converts a one-time occasion purchase into a potential platform relationship.

6. **Capitalize on the September–November demand peak with structured seasonal planning.** Given that revenue nearly doubles during this period, inventory availability, marketing spends, and customer acquisition campaigns should be front-loaded into Q3. Additionally, customers acquired during peak season should receive specific post-season retention campaigns in December–January to prevent the sharp drop-off that typically follows seasonal spikes.